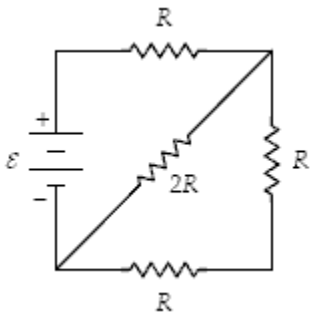
Physics 210 Tutorial #8 Multiple Choice Problems

____ 1. If 5.0×10^{21} electrons pass through a 20-Ω resistor in 10 min, what is the potential difference across the resistor?
   a. 21 V
   b. 32 V
   c. 27 V
   d. 37 V
   e. 54 V

____ 2. The temperature coefficient of resistivity of iron is 5.0×10^{-3} / °C; that of carbon is -0.50×10^{-3} / °C. When an iron wire and a carbon rod, each having the same 10 Ω resistance at 20°C, are cooled from that temperature to –80°C, the new ratio of the resistance of the carbon rod to the resistance of the iron wire at the lower temperature is
   a. –0.10.
   b. +1.9.
   c. +2.1.
   d. –10.
   e. +10.

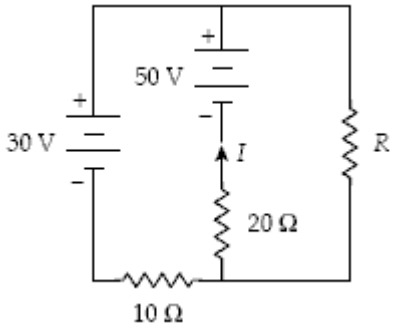
____ 3. Light bulb A is rated at 60 W and light bulb B is rated at 100 W. Both are designed to operate at 110 V. Which statement is correct?
   a. The 60 W bulb has a greater resistance and greater current than the 100 W bulb.
   b. The 60 W bulb has a greater resistance and smaller current than the 100 W bulb.
   c. The 60 W bulb has a smaller resistance and smaller current than the 100 W bulb.
   d. The 60 W bulb has a smaller resistance and greater current than the 100 W bulb.
   e. We need to know the resistivities of the filaments to answer this question.

____ 4. At what rate is thermal energy being generated in the 2*R*-resistor when $\varepsilon = 12$ V and $R = 3.0$ Ω?



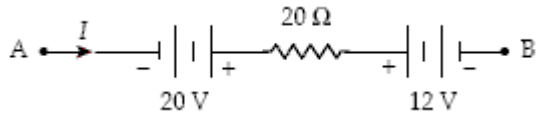
   a. 12 W
   b. 24 W
   c. 6.0 W
   d. 3.0 W
   e. 1.5 W

____ 5. Determine the magnitude and sense (direction) of the current in the 10-Ω resistor when I = 1.8 A.



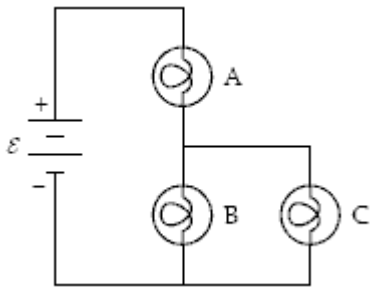
a. 1.6 A right to left
b. 1.6 A left to right
c. 1.2 A right to left
d. 1.2 A left to right
e. 1.8 A left to right

____ 6. What is the potential difference $V_B - V_A$ when the I = 1.5 A in the circuit segment below?



a. +22 V
b. –22 V
c. –38 V
d. +38 V
e. +2.0 V

____ 7. The circuit below contains three 100 watt light bulbs. The emf ε = 110 V. Which light bulb(s) is(are) the brightest?



a. A
b. B
c. C
d. B and C
e. All three are equally bright.

____ 8. An electron has a velocity of 6.0×10^6 m/s in the positive x direction at a point where the magnetic field has the components $B_x = 3.0$ T, $B_y = 1.5$ T, and $B_z = 2.0$ T. What is the magnitude of the acceleration of the electron at this point?
   a. 2.1×10^{18} m/s^2
   b. 1.6×10^{18} m/s^2
   c. 2.6×10^{18} m/s^2
   d. 3.2×10^{18} m/s^2
   e. 3.7×10^{18} m/s^2

____ 9. An electron moving in the positive x direction experiences a magnetic force in the positive z direction. If $B_x = 0$, what is the direction of the magnetic field?
   a. negative y direction
   b. positive y direction
   c. negative z direction
   d. positive z direction
   e. negative x direction

____ 10. A 2.0-C charge moves with a velocity of (2.0**i** + 4.0**j** + 6.0**k**) m/s and experiences a magnetic force of (4.0**i** – 20**j** + 12**k**) N. The x component of the magnetic field is equal to zero. Determine the y component of the magnetic field.
   a. –3.0 T
   b. +3.0 T
   c. +5.0 T
   d. –5.0 T
   e. +6.0 T

Physics 210 Multiple Choice Problems for week 7
Answer Section

MULTIPLE CHOICE

1. ANS: C PTS: 1
2. ANS: C PTS: 1
3. ANS: B PTS: 1
4. ANS: C PTS: 1
5. ANS: A PTS: 1
6. ANS: B PTS: 1
7. ANS: A PTS: 1
8. ANS: C PTS: 1
9. ANS: A PTS: 1
10. ANS: B PTS: 1